



STRATEGY & OPERATIONS

Vytalix Group Technology Architecture

Vytalix company-build documentation · reference
07_TECHNOLOGY_ARCHITECTURE

Status labels apply throughout: KNOWN, VERIFIED, ASSUMPTION, ESTIMATE, TO VALIDATE, PROPOSED. Nothing in this document claims traction, approvals, partners or revenue that do not exist. Governing file: FACTS_BASE.

PREPARED FOR

The Founder, Vytalix

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Owner: CTO, Vytalix · **Status:** PROPOSED v0.1 · **Date:** 7 September 2026 · **Governing file:** /FACTS_BASE.md .

Related: /docs/01_BUSINESS_MODEL.md (revenue streams that each component serves),
/docs/03_BRAND_SYSTEM.md (web and content standards), /docs/04_MATERNALINK_PRODUCT_STRATEGY.md ,
/product/MATERNALINK_MVP_SPEC.md (MaternaLink detail — not repeated here).

Starting point (VERIFIED): Vytalix has no domain, website, e-mail domain, CRM, LMS, cloud account, security tooling or company registration; the founder uses a personal Gmail; a MaternaLink prototype of unknown provenance runs on Vercel (FACTS_BASE §1). An investor CRM exists only as CSVs and a Python build script in this repository (/investors/). Everything below is a recommendation. All prices are ESTIMATE in GBP at September 2026 from public list prices as recalled — TO VALIDATE before purchase; vendor plan names change often.

Executive view (5 lines)

1. Vytalix needs two very different technology estates: a **lean, mostly bought, business stack** for the services and learning businesses that must cost under ~£300/month before MaternaLink build starts, and a **built, assured, in-region product platform** for MaternaLink (in development) that costs £350–£700/month to run at MVP plus the build itself.
2. **Buy, don't build**, for everything that is not the product: website (Webflow now, Next.js later), CRM (HubSpot Starter), LMS (LearnWorlds or Thinkific; Moodle only if a university/NHS buyer demands SCORM/LTI at scale), payments (Stripe + GoCardless; Paystack/Flutterwave for Nigeria), finance (Xero), workspace (Google Workspace).
3. **One product platform, not several:** MaternaLink's identity, audit, encryption, analytics and channel abstraction become the group's reusable core; the customer portal and admin dashboard are thin layers on it and are not built in year 1.
4. **AI is used for internal productivity and content drafting under a written policy:** no patient data to any third-party model without a DPIA, contractual no-training terms and an in-region endpoint; nothing generated reaches a woman without clinical sign-off.
5. **Assurance before scale:** Cyber Essentials (then Plus), DSPT, ICO registration, a fractional DPO and a Clinical Safety Officer are the year-1 governance spend; ISO 27001, FHIR/NHS API integration, native apps, a data warehouse and in-country Nigeria hosting are deliberately deferred.

1. Architecture diagram (group)

flowchart TB

```

subgraph Public[Public and marketing]
  WEB[Corporate website<br/>Webflow now -> Next.js + headless CMS later]
  LMS[E-learning platform<br/>LearnWorlds / Thinkific]
  AN1[Web analytics<br/>Plausible]
end

subgraph Business[Business operations - SaaS]
  GW[Google Workspace<br/>mail, docs, SS0]
  CRM[CRM<br/>HubSpot Starter]
  FIN[Finance<br/>Xero + Stripe + GoCardless]
  PAYNG[Nigeria payments<br/>Paystack / Flutterwave]
  DOCS[Contracts & docs<br/>PandaDoc / Docusign]
  PM[Work management<br/>Notion + GitHub Projects]
  SEC[Security tooling<br/>1Password, MDM, Cyber Essentials]
end

subgraph Product[MaternaLink platform - AWS eu-west-2, built]
  PWA[Woman & staff PWAs]
  API[NestJS API + workers]
  KC[Keycloak]
  PG[(PostgreSQL)]
  S3[(S3)]
  MB[Metabase dashboards]
  CH[Channel adapters<br/>Africa's Talking / Termii / Twilio / WhatsApp BSP]
end

subgraph Later[Deferred - v2+]
  PORTAL[Customer portal]
  FHIR[FHIR UK Core facade / NHS APIs - TO VALIDATE]
  DHIS[DHIS2 export]
  NGCELL[Nigeria hosting cell]
  DW[Data warehouse]
end

subgraph AI[AI - policy-governed]
  LLM[LLM tools for staff<br/>no patient data]
end

WEB --> CRM
LMS --> FIN
LMS --> CRM
WEB --> AN1
GW --> CRM
GW --> SEC
PWA --> API --> PG
API --> S3
API --> KC
API --> CH
PG --> MB
API -. -> FHIR
API -. -> DHIS
API -. -> NGCELL
PG -. -> DW
KC -. -> PORTAL
LLM -. -> WEB
LLM -. -> LMS

```

2. Component-by-component recommendations

Scoring: **Complexity** (L/M/H to set up and run), **Scalability** (L/M/H headroom without re-platforming).

2.1 Corporate website

Item	Recommendation
Technology	Year 1: Webflow (CMS plan) on <code>vytalix.com</code> or the cleared domain (name clearance TO VALIDATE — FACTS_BASE §1 conflicts). When engineers exist (2027+): Next.js static site + headless CMS (Sanity or Payload) deployed on Vercel or Cloudflare Pages
Why	No engineering time exists in year 1; the brand system (<code>/docs/03_BRAND_SYSTEM.md</code>) needs editable pages, a blog, case-study templates and forms; Webflow supports design fidelity and non-developer editing. Moving to Next.js later unifies the stack with the product
Cost (ESTIMATE)	Webflow CMS £18–£25/month; domain £10–£15/year; Next.js route £0–£20/month hosting + CMS £0–£80/month
Complexity / Scalability	L / M (Webflow); M / H (Next.js)
Alternatives	Framer (similar; cheaper design-led); WordPress (cheap, plugin/security burden); Squarespace (limited)
Requirements	WCAG 2.2 AA; cookie-less analytics; privacy notice; no claims beyond FACTS_BASE §4; HSTS; forms post to CRM

2.2 CMS

Item	Recommendation
Technology	Webflow CMS (year 1); Sanity (hosted, generous free tier, structured content) or Payload (open source, self-hostable in AWS) when the site moves to Next.js. MaternaLink's clinical content library is not a marketing CMS — it lives in the product with its own two-person approval workflow (PRD §9)
Why	Separate marketing content from clinical content; clinical content needs versioning, sign-off and audit
Cost (ESTIMATE)	Included in Webflow; Sanity £0–£80/month; Payload compute £20–£40/month
Complexity / Scalability	L / M; M / H
Alternatives	Contentful (pricier); Strapi (self-host)

2.3 CRM

Item	Recommendation
Technology	HubSpot (free CRM → Starter Customer Platform) for sales pipeline, contacts, sequences and marketing e-mail; the investor CRM in <code>/investors/</code> imports as a separate pipeline; Kemek Enterprise data stays out (FACTS_BASE §1)
Why	Fastest setup; free tier adequate for month 1–6; good Gmail integration; forms from Webflow; reporting for the sales engine in <code>/docs/13_SALES_ENGINE.md</code>
Cost (ESTIMATE)	£0 (free) → £15–£20/seat/month Starter; Professional £700+/month is not needed in years 1–2
Complexity / Scalability	L / H
Alternatives	Pipedrive (£12–£14/seat; sales-only, simpler); Attio (modern, £25+/seat); Folk (light); Salesforce (no)

Item	Recommendation
Data rules	Only business contact data; investor data per FACTS_BASE \$4 (public sources only); UK GDPR legitimate-interest assessment for outreach; suppression list

2.4 Customer portal

Item	Recommendation
Technology	Do not build in year 1. Serve consulting clients with shared Google Drive/Notion spaces and PandaDoc; serve e-learning customers inside the LMS; serve MaternaLink organisations inside the product's admin surface. Year 2–3: a Next.js portal on the MaternaLink identity (Keycloak) exposing invoices, documents, support tickets and product admin
Why	A portal without customers is cost; the product platform will supply identity and audit when needed
Cost (ESTIMATE)	£0 now; £30k–£60k build later
Complexity / Scalability	— / H (on the product platform)
Alternatives	HubSpot customer portal (ticketing); SuiteDash-style SaaS portals (weak security posture — TO VALIDATE)

2.5 E-learning platform

Serves revenue streams #9–#11 in `/docs/01_BUSINESS_MODEL.md` (courses £49–£299; subscriptions £19/month; corporate licences). Requirements: self-paced video/text courses, quizzes, certificates (no accreditation claims until CPD approval — TO VALIDATE), coupons and regional pricing (Nigeria/Kenya/Ghana), bundles/subscriptions, corporate seat licences, SCORM export/import if NHS/universities require, GDPR-compliant learner data, UK VAT handling.

Option	Fit	Cost (ESTIMATE, monthly)	Complexity	Scalability	Notes
LearnWorlds (recommended for B2B + B2C)	Strong: interactive video, assessments, certificates, SCORM, white-label, corporate/team plans, mobile app add-on	£25–£250 (Starter to Learning Center); transaction fee on the lowest tier	L–M	H	Best match for corporate licences and NHS-style buyers; TO VALIDATE current plan names and fees
Thinkific (alternative, cheapest credible start)	Good: simple authoring, communities, no transaction fees on paid plans (TO VALIDATE), B2B "groups"	£30–£150	L	H	Choose if year-1 focus is B2C courses only
Teachable	Good for B2C creators; payment handling included; less B2B	£30–£120 + fees on lower tiers	L	M	Weaker corporate/SCORM story
Moodle (self-hosted or MoodleCloud)	Full LMS, SCORM/LTI, accreditation-friendly, open source	MoodleCloud £80–£150; self-host £30–£100 compute + admin time	M–H	H	Choose only when a university/NHS licence demands LTI/SCORM at scale or data must be self-hosted; admin burden for a one-person company

Option	Fit	Cost (ESTIMATE, monthly)	Complexity	Scalability	Notes
Custom (Next.js + Stripe + video CDN)	Full control	£40k–£80k build + £100+/month	H	H	Not before year 3; no justification at current volumes

Recommendation: start on **Thinkific** if the first courses are founder-authored B2C (months 6–9), or **LearnWorlds** if a corporate/NHS cohort licence is signed first; migrate to Moodle only on a buyer requirement. Video hosting via the LMS (or Vimeo/Bunny) and course content drafted with AI assistance under \$2.13.

2.6 MaternaLink application

See [/product/MATERNALINK_MVP_SPEC.md](#) for the full stack. Summary for the group view: TypeScript monorepo; Next.js PWAs; NestJS API; PostgreSQL (RLS) on RDS; Redis/BullMQ; S3; Keycloak; AWS eu-west-2; Terraform; GitHub Actions; channel adapters (Africa's Talking/Termii/Twilio; WhatsApp BSP v1.1). Run cost £350–£700/month at MVP; build £140k–£320k (ESTIMATE). Complexity H; Scalability H (designed to 20k women / 2m SMS per year before change).

2.7 Admin dashboard

Item	Recommendation
Technology	Product admin inside the MaternaLink staff app (organisations, users, content workflow, audit viewer, feature flags) — part of the MVP; business dashboards on Metabase (self-hosted on ECS) reading a read replica; group KPI dashboard (services revenue, pipeline, learners) initially a Google Sheet/Looker Studio fed by HubSpot, Xero and LMS exports
Why	Avoids a separate internal-tools platform; Metabase is open source and embeds for <code>programme_manager</code> views
Cost (ESTIMATE)	Metabase compute £30–£50/month; Looker Studio £0
Complexity / Scalability	M / H
Alternatives	Retool (fast internal tools; £10+/user; data leaves region unless self-hosted); Grafana for ops; Power BI if the buyer base is Microsoft-heavy

2.8 Analytics

Layer	Recommendation	Cost (ESTIMATE)	Alternatives
Web analytics	Plausible (EU-hosted, cookie-less) on website and LMS	£9–£15/month	Fathom; GA4 (free; consent banner and data-transfer questions)
Product analytics (MaternaLink)	In-house events table (no PII) → Metabase; PostHog EU self-hosted considered in year 2	Included in cloud	Amplitude/Mixpanel (data leaves region — excluded for patient-adjacent data)
Business KPIs	Looker Studio / Sheets over HubSpot, Xero, LMS	£0	Metabase over exported CSVs
Warehouse	Not in year 1 ; Parquet on S3 + Athena when volumes justify; BigQuery/Snowflake later	£0 → £50–£200/month later	—

2.9 Authentication and identity

Item	Recommendation
Staff/business tools	Google Workspace as the identity provider (SSO to HubSpot, Notion, GitHub, LMS admin); mandatory 2-step verification with security keys/passkeys for admins; a company domain replaces personal Gmail on day one of incorporation
MaternaLink staff users	Keycloak (self-hosted, eu-west-2) — OIDC for our apps, SAML/OIDC federation with NHS trust or state identity providers; MFA enforced
MaternaLink women	Phone OTP + device PIN/biometric (MVP spec §2)
Customer portal (later)	Keycloak realm per customer type
Cost (ESTIMATE)	Workspace £5–£11/user/month; Keycloak £40–£70/month compute
Alternatives	Auth0 (managed; per-MAU; UK residency tier TO VALIDATE); Clerk (developer-friendly; residency TO VALIDATE); Cognito (cheap; weaker federation UX); Microsoft Entra ID if the company standardises on Microsoft 365 (NHS buyers are Microsoft-heavy — a legitimate reason to choose Microsoft 365 over Google Workspace; decision TO VALIDATE with the founder's preference)

2.10 Payments and billing

Need	Recommendation	Fees (ESTIMATE, TO VALIDATE)	Alternatives
Course and subscription payments (UK/global cards)	Stripe (Checkout, Billing, Tax for VAT on digital services) — usually via the LMS's built-in Stripe integration	~1.5% + 20p UK cards; ~2.5%+ international; Stripe Tax extra	Paddle (merchant of record — handles global VAT; higher %); Lemon Squeezy
Retainers and invoices (UK B2B)	Xero invoicing + GoCardless direct debit for retainers; Stripe for card-paid invoices	GoCardless ~1% + 20p capped per transaction (TO VALIDATE); Xero £15–£40/month	QuickBooks; FreeAgent
Nigeria (courses, programme fees, pilots)	Paystack (Stripe-owned) or Flutterwave for NGN cards, bank transfer, USSD and mobile money; requires a Nigerian entity/bank account or a partner (TO VALIDATE)	~1.5% local + ₦100 on transactions over ₦2,500 (Paystack, as recalled — TO VALIDATE); Flutterwave similar; international cards ~3.9%	Interswitch; Monnify; bank transfer with manual reconciliation
Government/donor contracts	Invoice + bank transfer; no card rails; consider escrow/milestone payments and FX policy	Bank fees	—
Rules	No card data touches Vytalix systems (PCI SAQ-A via hosted checkout); refunds per Consumer Rights Act 2015; SMS pass-through billed monthly from <code>delivery_attempt</code> cost records		

2.11 APIs and integration layer

Item	Recommendation
Product APIs	OpenAPI 3.1 REST (MVP spec §4); versioned; per-organisation API keys for programme partners (v1.2) with scopes; webhooks for events (v2)
Health interoperability	FHIR UK Core facade (read first) in v2; PRSB digital maternity record standard alignment for the handover summary (TO VALIDATE current version); HL7 UK membership when integration starts

Item	Recommendation
NHS APIs (all TO VALIDATE for access, onboarding and fit)	PDS FHIR API (demographics), NHS login (citizen identity), MESH (secure messaging), GP Connect (not maternity), NHS App integration (in-app messaging/notifications), Maternity Services Data Set (MSDS) alignment for indicators — each requires onboarding through the NHS England API platform with a clinical safety and IG case
Nigeria	DHIS2 aggregate export (v1.2) mapped to the state's data elements; NHIA/state insurance interfaces TO VALIDATE
EPR vendors	Vendor-supported export/import interfaces (BadgerNet, Euroking, K2 — availability TO VALIDATE) in v2.2
iPaaS	Not needed; point integrations via the API and workers; Zapier/Make only for business tools (HubSpot ↔ Xero ↔ LMS), never for patient data
Cost (ESTIMATE)	NHS API onboarding is mostly effort (£20k–£40k engineering and assurance per integration); FHIR facade £30k–£50k build

2.12 Cloud, databases and hosting

Item	Recommendation
Primary cloud	AWS eu-west-2 (London) with separate accounts for prod, non-prod, backup vault and security tooling under AWS Organizations; SSO via Workspace/IAM Identity Center
Alternative	Azure UK South — equivalent services; choose if the first NHS customer or the founder's team is Microsoft-centred. Do not run both
Nigeria	Cell options A–D in MVP spec §2.1; default: UK region with NDPA transfer safeguards until a contract requires in-country hosting
Databases	PostgreSQL everywhere (RDS); no MongoDB/Firebase for personal data (weaker relational integrity and audit); Redis for queues/cache only
Object storage	S3 (SSE-KMS, versioning, Object Lock for audit)
Edge	CloudFront + WAF; Cloudflare as DNS/registrar alternative
Cost (ESTIMATE)	£350–£700/month at MVP; £1,000–£2,500/month at 5 organisations and 20k women (ESTIMATE); Nigeria cell +£350–£1,500/month
Complexity / Scalability	H / H
Alternatives	GCP europe-west2 (fewer NHS references — ASSUMPTION); Hetzner/OVH (cheap, weak compliance story for NHS/DSPT); Vercel/Supabase (excellent DX; sub-processor and residency questions for patient data — acceptable for non-clinical apps such as the corporate site)

2.13 AI and LLM usage policy (group-wide, PROPOSED)

Rule	Detail
Permitted now	Drafting marketing and course content, code assistance, meeting summaries of internal meetings, research synthesis, translation drafts for internal review — using enterprise/business tiers with no-training terms (e.g., business tiers of major LLM vendors; TO VALIDATE terms and regions)
Prohibited	Entering any patient/woman data, identifiable client data, unpublished clinical content without review, or confidential contract terms into any third-party model without a DPIA and a contract covering the processing; generating clinical advice shown to women; using LLM output as a clinical or regulatory decision

Rule	Detail
MaternaLink product use	v1: none on patient data. v1.1–v2: operational message-type routing (not urgency), staff-authored translation drafts, content variants — all human-reviewed, in a UK/EU-region endpoint under DPIA and no-training contract (strategy §16). v3: research only under ethics approvals
Governance	AI use register; model/vendor list approved by the CTO; prompts and outputs for regulated content stored with the content version; bias and safety review for any woman-facing content pipeline; annual policy review
Tooling cost (ESTIMATE)	£20–£60/user/month for assistant tools; coding assistants £10–£40/user/month

2.14 Cybersecurity (group)

Control	Recommendation	Cost (ESTIMATE)
Identity	Workspace SSO, passkeys/security keys for admins, no shared accounts	Included
Devices	MDM (Google endpoint management for basics; Kandji/Jamf for Mac or Intune for Windows when >5 devices); disk encryption; auto-updates; EDR (e.g., CrowdStrike/Defender for Business) once staff >5	£0–£8/device/month
Passwords/secrets	1Password Business (or Bitwarden) for the team; AWS Secrets Manager for systems	£6–£8/user/month
E-mail security	SPF/DKIM/DMARC (p=reject), phishing-resistant MFA, security awareness training	£0–£5/user/month
Certification	Cyber Essentials (basic £300–£600 + VAT by size band; Plus £1,500–£3,000 + VAT — sources: https://www.isms.online/cyber-essentials/cost/ ; https://www.figggroup.co.uk/blog/cyber-essentials-vs-cyber-essentials-plus-cost/); DSPT as a Category 3 supplier (v8 CAF-aligned: 35 assertions, 42 evidence items — https://www.dsptoolkit.nhs.uk/News/161); ISO 27001 deferred to year 2–3 (£15k–£40k)	£2k–£4k/year
Product security	See MVP spec §12 (WAF, scanning, pen test, audit, encryption)	£6k–£12k/year pen test
Insurance	Cyber and professional indemnity once trading (TO VALIDATE quotes)	£1k–£3k/year
Incident response	Written plan; 72-hour ICO/NDPC notification readiness; tabletop exercise twice a year	Effort

2.15 Data governance

Element	Recommendation
Registrations	ICO registration on incorporation (fee tier ESTIMATE £52–£78 — TO VALIDATE); NDPC registration when Nigeria thresholds apply (health sector; reportedly >200 data subjects in 6 months — TO VALIDATE against https://ndpc.gov.ng/wp-content/uploads/2025/07/NDP-ACT-GAID-2025-MARCH-20TH.pdf)
Roles	Fractional DPO (£500–£1,500/month ESTIMATE); SIRO (founder); Caldicott-style guardian from the clinical advisory group; CSO (DCB0129)
Artefacts	RoPA; data classification (public / internal / confidential / special category); retention schedule; DPIA templates (product, marketing, LMS); sub-processor register; DPAs; international transfer assessments; privacy notices per audience and language
Data classes and where they live	Marketing/CRM (HubSpot, EU/US hosting TO VALIDATE → transfer assessment); learners (LMS); finance (Xero/Stripe); patient-adjacent (MaternaLink only, in-region) — never in Drive/Notion/CRM
Access	Least privilege; quarterly access reviews; joiner/leaver checklist
Anonymisation	Aggregate dashboards with small-cell suppression; ICO anonymisation guidance

2.16 Backups and continuity

Layer	Recommendation	Cost (ESTIMATE)
Product	MVP spec §9: PITR, 35-day snapshots, cross-account UK copy, RPO 15 min / RTO 4 h, quarterly restore drills	Included in cloud
Business SaaS	Google Workspace backup (e.g., Spanning/Backupify-class) once >3 users; HubSpot/Xero exports monthly to an encrypted S3 bucket; LMS content master copies in Drive + S3	£3–£6/user/month
Code	GitHub organisation with branch protection; nightly mirror to S3	£0–£10/month
Continuity	Static status/emergency page; documented manual fallback for MaternaLink sites (phone lists)	Effort

3. Lean year-1 stack and monthly total (ESTIMATE)

Two phases: **Phase A** (months 1–3, services only) and **Phase B** (months 4–12, MaternaLink build and alpha).

ASSUMPTION: 2 paid seats in Phase A, 4–5 in Phase B.

Component	Phase A (£/month)	Phase B (£/month)	Notes
Domain, DNS, e-mail security	2	2	Cloudflare registrar; DMARC tooling free tier
Google Workspace (Business Starter/Standard)	10–22	25–55	Replaces personal Gmail
Website (Webflow CMS)	18–25	18–25	
Web analytics (Plausible)	9	9	
CRM (HubSpot free → Starter)	0–20	20–40	
Finance (Xero) + GoCardless/Stripe	15–30	15–30	Transaction fees excluded
Work management (Notion/GitHub Projects)	0–16	16–40	
Contracts (PandaDoc/Docusign)	0–25	25–40	
Password manager (1Password)	12–16	30–40	
Scheduling (Calendly)	0–10	0–10	
AI assistants (business tier)	20–60	80–200	Per policy \$2.13
LMS (Thinkific/LearnWorlds)	0	30–120	From month ~6
GitHub Team	0	12–20	
AWS (dev/staging/prod as they come online)	0	150–700	Ramps across the build
Observability (Sentry, Grafana Cloud, uptime)	0	20–80	
Messaging gateways (sandbox → pilot)	0	10–150	Pass-through in pilots
Fractional DPO	0–500	500–1,500	Governance, not tooling; shown for completeness
Tooling total (excluding DPO)	£90–£250	£460–£1,560	

Component	Phase A (£/month)	Phase B (£/month)	Notes
Total including DPO	£90–£750	£960–£3,060	

Annual items (ESTIMATE): Cyber Essentials £300–£600; Plus £1,500–£3,000; pen test £6k–£12k; accessibility audit £3k–£6k; ICO fee £52–£78; insurance £1k–£3k.

4. What not to build (yet)

Item	Why not now	Revisit when
Custom website/CMS	No engineers; Webflow suffices	Engineering team in place and brand assets stable
Custom LMS	Off-the-shelf covers years 1–2	>£150k/year learning revenue or a buyer demands self-hosting
Customer portal	No customers to serve	5+ paying organisations
Data warehouse / BI platform	Volumes tiny; Metabase and Sheets suffice	>5 organisations or research data needs
Native mobile apps	PWA meets MVP needs; app-store overhead and clinical-safety re-testing per release	Evaluation shows PWA limits (push on iOS, background sync)
FHIR / NHS API integrations	No NHS customer; onboarding is effort-heavy	A trust asks for it (v2)
Nigeria in-country hosting	No contract requiring it	A signed contract with a residency clause
Multi-region active-active	Cost and complexity	>99.9% requirement in a contract
AI features on patient data	No DPIA, no in-region endpoint, no clinical review process	v1.1+ under strategy §16 with the policy in §2.13
ISO 27001	Cyber Essentials Plus + DSPT are what NHS buyers ask first	A buyer requires it or year 3
Kubernetes	ECS Fargate is enough	Multi-service scale or team preference with ops capacity
Microservices	Modular monolith (NestJS modules) is faster and safer	Team >10 engineers
Separate identity SaaS for women	OTP+PIN suffices	Unification with Keycloak justified
Blockchain, tokenisation, wearables	No user need; distracts from the credibility strategy	Never, absent evidence

5. Integration and data-flow principles

1. Patient-adjacent data lives only in the MaternaLink platform, in-region, behind RLS, audit and encryption; business SaaS never receives it.
2. Every external service is a sub-processor with a DPA and an entry in the register; every new one requires a DPIA delta.

- Integrations are adapters behind interfaces (channel providers, identity providers, EPR exports, DHIS2) so that a customer choice is configuration.
- Exports for customers are documented formats (CSV, FHIR bundles v2, DHIS2 aggregates) to avoid lock-in.
- Observability, backups and IaC are non-negotiable from the first environment — they are the cheapest DSPT evidence.

Priorities / Risks / Next actions

Priorities

- On incorporation: domain and Google Workspace (or Microsoft 365 — decide once), DMARC, 1Password, HubSpot free, Xero, Webflow site, Plausible, ICO registration — one week of setup, under £300/month.
- Stand up the AWS organisation, Terraform baseline and GitHub before the MaternaLink build starts; adopt the LLM policy (§2.13) in writing.
- Choose the LMS at the point the first course is scheduled (month 6), not before.
- Book Cyber Essentials for the quarter in which MaternaLink staging exists; plan DSPT for the cycle in which real patient data first flows.
- Keep the deferred list (§4) as a standing agenda item so that build pressure does not reintroduce it early.

Risks

Risk	Mitigation
Founder time consumed by tool sprawl	Single-week setup checklist; SaaS with SSO only; quarterly tool review
Personal Gmail and Drive continue to hold company data	Migrate on day one of the domain; Drive ownership transfer; Kemek separation
Vendor data residency assumptions wrong (HubSpot, Clerk, LMS)	Transfer assessments; keep patient data out of SaaS entirely
Cloud costs creep during build	Budgets and alerts; Fargate Spot in non-prod; monthly review
Choosing Google Workspace when NHS buyers live in Microsoft	Decide once with the founder; either works — federation is what matters
Prototype on Vercel becomes a shadow production system	Audit and freeze it (strategy §25); no real data

Next actions

- CTO: setup checklist and IaC baseline; sub-processor register v0.1; AI policy v0.1 for founder sign-off.
- Founder: domain and name clearance; workspace decision (Google vs Microsoft); ICO registration on incorporation; Vercel prototype access and audit authorisation.
- Finance/ops (founder or bookkeeper): Xero, Stripe, GoCardless accounts; Paystack/Flutterwave entity question for Nigeria (TO VALIDATE).
- DPO (fractional, to contract): RoPA, DPIA templates, retention schedule.